Linear Algebra Lecture Notes

Chapter 5

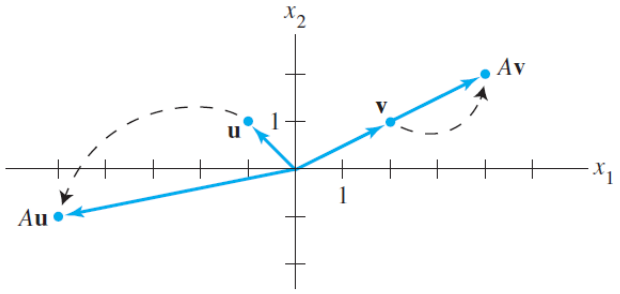
5.1 Eigenvectors and eigenvalues

– It often happens that there are special vectors on which the action of matrix A is quite simple.

Ex: Let $A = \begin{bmatrix} 1 & -2 \\ 1 & 0 \end{bmatrix}$, $u = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ $v = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$.

Images of u and v under multiplication by A are as follows:

$Au = \begin{bmatrix} -5 \\ -1 \end{bmatrix}$ | $Av = \begin{bmatrix} 4 \\ 2 \end{bmatrix} = 2v$



– This section studies equations such as $Ax = 2x$ or $Ax = -4x$ where special vectors are transformed by A into scalar multiples of themselves.

Definition :

$\lambda \in \mathbb{R}$ nonzero

An **eigenvector** of an $n \times n$ matrix A is a nonzero vector $\mathbf{x}$ such that $A\mathbf{x} = \lambda\mathbf{x}$ for some scalar λ. A scalar λ is called an **eigenvalue** of A if there is a nontrivial solution $\mathbf{x}$ of $A\mathbf{x} = \lambda\mathbf{x}$; such an $\mathbf{x}$ is called an *eigenvector corresponding to* λ.[1]

– Note that an eigenvector must be nonzero, by definition, but an eigenvalue may be zero. The case in which the number 0 is an eigenvalue is discussed in the following examples.

– It is easy to determine if a given vector is an eigenvector of a matrix. It is also easy to decide if a specified scalar is an eigenvalue.

Ex: Let $A = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$ $u = \begin{bmatrix} 6 \\ -5 \end{bmatrix}$, $v = \begin{bmatrix} 3 \\ -2 \end{bmatrix}$. ✓ ✗

Are u and v eigenvectors of A.

Sol.

$$A \cdot u = \begin{bmatrix} -24 \\ 20 \end{bmatrix} = (-4)\begin{bmatrix} 6 \\ -5 \end{bmatrix} = -4 \cdot u \quad ✓$$

$$A \cdot v = \begin{bmatrix} -9 \\ 11 \end{bmatrix} = (?)\begin{bmatrix} 3 \\ -2 \end{bmatrix} \quad ✗$$

Ex: Show that 7 is an eigenvalue of $A = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$ and find the corresponding eigenvectors.

Sol. $Ax = 7x$ must have a nontrivial solution, by definition.

Then $Ax - 7x = 0$ or $(A - 7I)x = 0$.

To solve this homogeneous equation

$$A - 7I = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} - \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix} = \begin{bmatrix} -6 & 6 \\ 5 & -5 \end{bmatrix}$$

Columns of $A - 7I$ are obviously linearly dependent. So $(A - 7I)x = 0$ has nontrivial solutions. Thus, 7 is an eigenvalue of A. To find corresponding eigenvectors use row operations:

$$\begin{bmatrix} -6 & 6 & 0 \\ 5 & -5 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

x_2 is free

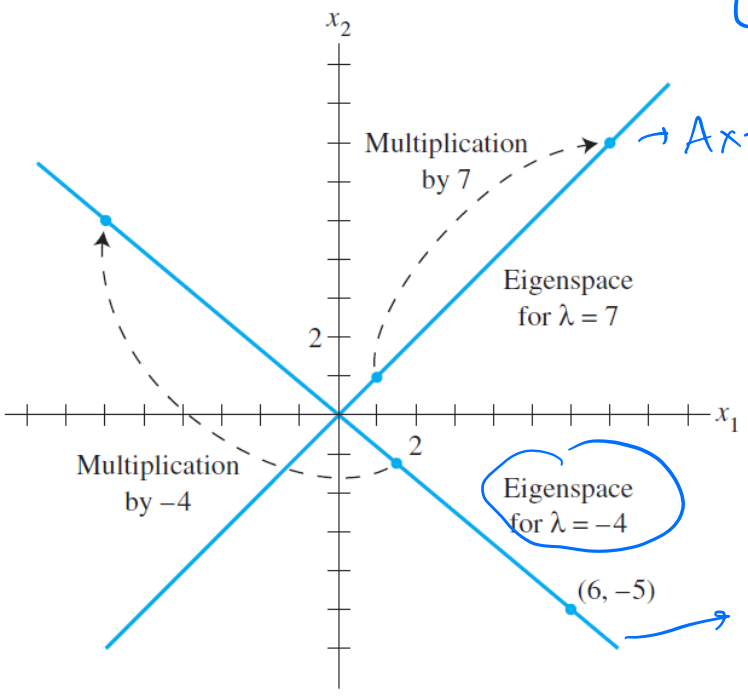
General solution is $x_1 - x_2 = 0 \Rightarrow x_1 = x_2$

$\Rightarrow \quad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

Each vector of this form ($x_2 \neq 0$) is an eigenvector corresponding to $\lambda = 7$. i.e. $\begin{bmatrix} 2 \\ 2 \end{bmatrix}, \begin{bmatrix} 7 \\ 7 \end{bmatrix}$ ...

- Note: Although row reduction was used in previous example to find eigenvectors, it can not be used to find eigenvalues.
- The set of all solutions of $(A - \lambda I)x = 0$ is just the null space of the matrix $A - \lambda I$. So, this set is a subspace of $\mathbb{R}^n$ and is called the eigenspace of A corresponding λ. The eigenspace consists of the zero vector and all the eigenvectors corresponding to λ.

$A = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$ $\lambda_1 = 7$ $\lambda_2 = -4$



$\rightarrow Ax = 7x$

$\rightarrow Ax = -4x$.

Ex: $A = \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix}$ An eigenvalue is 2. Find a basis for corresponding eigenspace. $Ax = 2x$?

Sol

$$A - 2I = \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix} - \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 2 & -1 & 6 \\ 2 & -1 & 6 \\ 2 & -1 & 6 \end{bmatrix}$$

Now reduce the augmented matrix for $(A - 2I)x = 0$

$$\begin{bmatrix} 2 & -1 & 6 & 0 \\ 2 & -1 & 6 & 0 \\ 2 & -1 & 6 & 0 \end{bmatrix} \sim \begin{bmatrix} 2 & -1 & 6 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow 2x_1 - x_2 + 6x_3 = 0$$

x_2, x_3 free

$$x_1 = \frac{x_2}{2} - 3x_3$$

$$\Rightarrow x = \begin{bmatrix} \frac{x_2}{2} - 3x_3 \\ x_2 \\ x_3 \end{bmatrix} \rightarrow x = x_2 \begin{bmatrix} 1/2 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}$$

The eigenspace is a two-dimensional subspace of $\mathbb{R}^3$.

A basis is $\left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \right\}$.

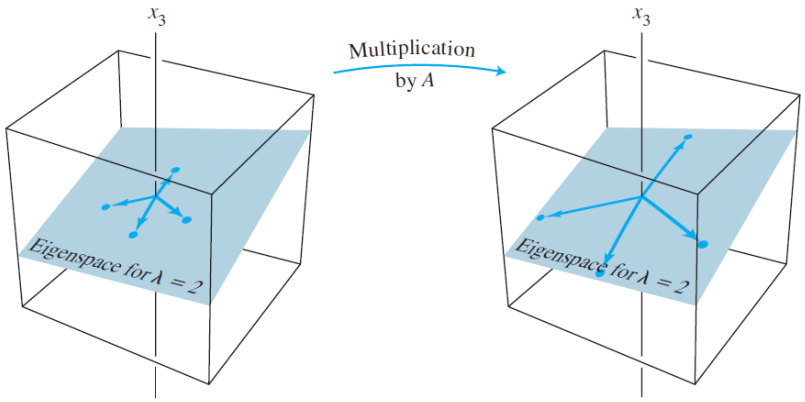
Check and see that a, b are in subspace generated by $\left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \right\}$

$$A \cdot \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} = a = \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 0 \end{bmatrix} = 2 \cdot \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$$

$$A \cdot \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} = b = \begin{bmatrix} -6 \\ 0 \\ 2 \end{bmatrix} = 2 \cdot \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}$$

$A \cdot \begin{bmatrix} -1 \\ 4 \\ 1 \end{bmatrix} = A(2a+b) = \begin{bmatrix} -2 \\ 8 \\ 2 \end{bmatrix} = 2(2a+b)$

$c = 2a+b$, so it is also in the subspace generated by $\left\{\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} -3 \\ 6 \\ 1 \end{bmatrix}\right\}$



Theorem: The eigenvalues of a triangular matrix are the entries on its main diagonal.

Ex: $A = \begin{bmatrix} 3 & 6 & -8 \\ 0 & 0 & 6 \\ 0 & 0 & 2 \end{bmatrix}$ $B = \begin{bmatrix} 4 & 0 & 0 \\ -2 & 1 & 0 \\ 5 & 3 & 4 \end{bmatrix}$

The eigenvalues of A are $3, 0, 2$

B : $4, 1$.

$Ax = \lambda x$

– What does it mean for a matrix A to have an eigenvalue of 0, such as the above example? This happens iff the equation $Ax = 0x$ has a nontrivial solution. But $Ax = 0x$ is equivalent to $Ax = 0$ which has a nontrivial solution iff A is not invertible. Thus 0 is an eigenvalue iff A is not invertible.

Theorem : If $v_1, v_2, \ldots, v_r$ are eigenvectors that correspond to distinct eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_r$ of an $n \times n$ matrix A, then the set $\{v_1, v_2, \ldots, v_r\}$ is linearly independent.